OpenRocket Extraterrestrial Flight Design Challenge

Team Name: TON618 **Rocket Body Material: Aluminium**

1. Design approach and thought process

Mission and first estimate. The rocket must carry a 1.00 kg payload to 4,000(+/-)100 m on Mustafar, stay under 2,800 N・s of total impulse and 2,200 mm in length, and recover the payload and rocket under parachute. Ignoring drag, reaching 4 km at g =6.5 m/s² needs a burnout speed of about √(2gh) ≈ 228 m/s, so with drag losses we expected peak speeds of 250–300 m/s.

OpenRocket gives a speed of sound of about 590 m/s at 700 K, so the whole flight stays subsonic (Mach < 0.55). Wave drag is therefore irrelevant, and the nose and fin choices were made for subsonic flight.

Choices and reasons:

• **Material:** aluminium for every structural part (nose, tubes, motor mounts, rings, fins, lugs), limit 15 kg. Launch mass is 4.46 kg, so the cap is not a constraint, and one material keeps the mass model simple and consistent with the rules.

• **Nose cone:** ogive, 300 mm long on a 75 mm body (fineness ratio 4). At Mach < 0.55 pressure drag dominate, so a smooth ogive is an efficient, easy-to-model choice that also gives room for the payload section.

• **Layout:** three sections on one 75 mm diameter. Stage 1 is the booster, stage 2 the sustainer, and stage 3 a payload section (nose, payload tube, 1.00 kg payload, drogue and payload parachute) that can separate during descent as the problem statement requires.

• **Motors (OpenRocket database, 38 mm):** booster Cesaroni 648J285-15A (648 N・s) and sustainer AeroTech J350W (≈666 N・s). Combined impulse is ≈1,314 N・s, 47 % of the 2,800 N・s limit. Impulse was calculated firstly using basic pen and paper calculations and then hit and trial on OpenRocket.

• **Staging:** There are mainly two stages:

1.Stage1(Booster) ->This stage is designed to provide initial thrust to rocket. Separated after burnout

2.Stage2(Sustainer)->Ignites after booster burnout and carries the rocket to a higher altitude. We have designed a separate stage for payload deployment also so that payload and sustainer can be recovered separately.

History.: We started from a small model rocket (A8/B6 motors, 21 m apogee) to confirm the Mustafar environment worked, then rescaled to a 75 mm body. Early runs with K-class motors exceeded the 2,800 N・s limit (≈3,200 N・s) and the sustainer tumbled after

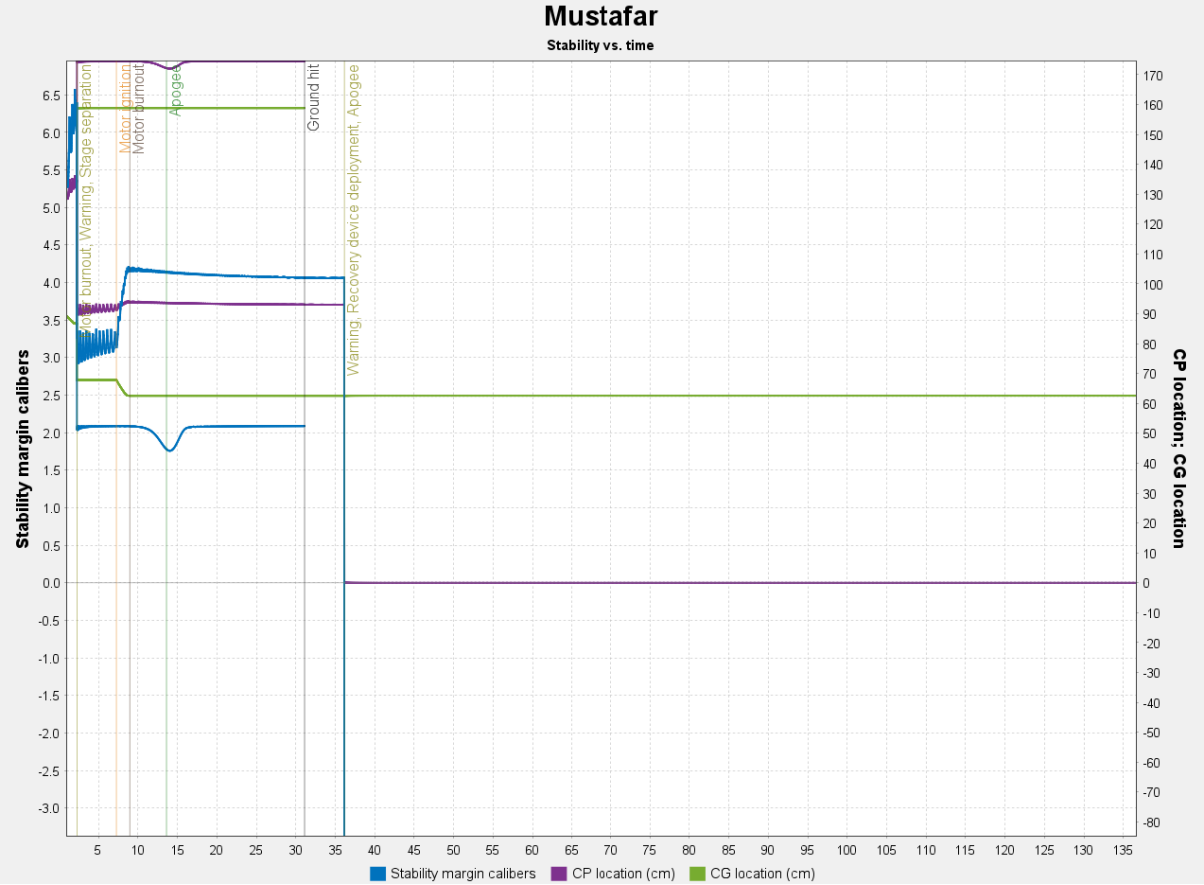
separation with the original small fins, so we moved to J-class motors and larger fins. We then tuned the wall thickness (mass), the sustainer ejection delay (27.2 s) and the parachute sizes, one change per run, re-checking apogee, stability, length and descent speed after each run.

2. Final design and simulated characteristics:

Section	Component	Dimensions	Description
Stage 3 (Payload)	Nose cone (ogive)	30cm,wall 0.2cm	Aluminium
	Payload tube	25 cm, Diameter 7.5 cm, wall 0.2 cm	1.00 kg payload (mass component, no override); drogue diameter 0.9 m and payload parachute diameter 1.5 m
Stage 2 (sustainer)	Sustainer tube	70cm ,wall0.125cm	J350W (3.8 cm), sustainer parachute diameter 1.3 m, launch lug
	Fins	4×, root 50, tip 50, height 60, sweep 25, thick 3 mm	Aluminium trapezoidal
Stage1 (Booster)	Booster tube	50 cm, wall 0.1 cm	648H285-15A (38 mm), launch lug
	Fins	4×, root 50, tip 50, height 80, sweep 25, thick 3 mm	Aluminium trapezoidal
Whole rocket	Length/diameter	178cm/7.5cm	Limit 220cm

Mass and propulsion(Mustafar Run)	Value
Launch mass (incl. both motors)	4.459 kg (aluminium cap 15 kg)
Mass after booster separation / at sustainer burnout	3.279 kg / 2.918 kg
Final masses: payload section / sustainer / booster	1.762 kg / 1.156 kg / 0.852 kg
Booster 648J285-15A: impulse, burn, peak thrust	648 N・s, 2.25 s, 369 N (loaded mass ≈0.60 kg)
Sustainer J350W: impulse, burn, peak thrust	≈666 N・s, 1.7 s, 720 N (loaded mass ≈0.65 kg)
Thrust-to-weight at liftoff; rod exit	≈12; 21.6 m/s at 0.32 s (3.0 m rod)
Maximum velocity; maximum Mach number	283 m/s; 0.51 (speed of sound 590 m/s)

3.Simulated stability Characteristics:



The rocket remains stable during the entire flight, but because it is overstable(about 3–6.6 cal), it turns too much into the wind. In a 10 m/s wind, it drifts about 1.8 km upwind and is still traveling sideways at around 43 m/s when the drogue parachute opens.

4. Recovery system:

Time	Altitude	Event
2.26 s	≈170 m	Booster burns out and separates; it falls without a parachute.
7.26–8.96 s		Sustainer ignites and burns out; the stack coasts to apogee.
36.1 s	4,010 m	Apogee. The 0.9 m drogue deploys; payload section and sustainer descend together .
183.8 s	2,000 m	Stage 3 separates from the sustainer. The diameter 1.5 m payload parachute and diameter 1.3 m sustainer

		parachute (6 shroud lines each, ripstop nylon) open
≈540 s	0 m	Final vertical speeds 5.65 m/s (payload section) and 6.16 m/s (sustainer).

NOTE:The ground-hit speed of 11.5 m/s reported by OpenRocket is the total speed including the 10 m/s sideways drift and the vertical descent rate.

5. Earth versus Mustafar: predictions and tests:

Method-> The same rocket, motors, rod (3.0 m) and recovery settings were flown in seven simulations. The Earth baseline uses the standard atmosphere (101.3 kPa, 15 °C, ρ = 1.225 kg/m ^{3}), g = 9.79 m/s and a 2 m/s wind (assumption). Each single-parameter run changes only one setting from this baseline. Density cannot be set directly in OpenRocket, so its effect appears through the pressure and temperature runs, and the "P + T" run gives the Mustafar density of 0.373 kg/m ^{3}.

Run	Change from earth	Density (kg/m ^{3})	Apogee (m)	Max Speed(m/s)	Max Mach	Stability on ascent(cal)	Decent speed (Payload / Sustainer)	Landing distance(m) (Payload/Sustainer)
1 Earth baseline	-------	1.225	2120	232	0.69	3.35-6.61	3.83/4.17	639/557
2 Gravity	g=6.5m/s ^{2}	1.225	2687	250	0.74	2.67-6.61	3.12/3.40	1082/982
3 Pressure	75kPa	0.907	2415	241	0.72	3.29-6.62	4.45/4.85	529/458
4 Temperature	700K	0.504	2915	255	0.45	3.11-6.55	5.97/6.50	383/327
5 Wind	10m/s	1.225	1648	232	0.70	3.20-6.61	9.58/N.A	331/N.A
6Pressure+temp	75 kPa, 700K	0.373	3210	261	0.46	3.00-6.57	6.93/7.56	285/237
7 Full Mustafar	All of the above	0.373	4010	283	0.51	2.93-6.57	5.65/6.16	3190/2896

NOTE:Run 5 apogee (1,648 m) is below the 2,000 m trigger, so the main chutes never deployed; the 9.58 m/s is the drogue-only descent and the run ended after the payload section landed.

Hypotheses and outcomes:

Change	Prediction	Result	True?(prediction)

Lower gravity	Less weight and slower deceleration, so a higher apogee and a longer climb.	Apogee 2,120 → 2,687 m (+27 %); time to apogee 23.4 → 29.1 s (+24 %); max speed 232 → 250 m/s.	Yes
Thinner Air	Less drag, so higher apogee and speed, but faster parachute descent and some change in stability.	Apogee +14 %, max speed +4 %; payload descent 3.83 → 4.45 m/s (+16 %). Stability range barely changed (minimum 3.35 → 3.29 cal).	Yes for drag and descent; no for stability (negligible)
Hotter air	Higher speed of sound, so a lower Mach number for the same speed.	Speed of sound 340 → 590 m/s; max Mach 0.69 → 0.45.	Yes
10 m/s side wind (from 2 m/s)	Weathercocking into the wind, lost altitude and sideways drift.	Apogee 2,120 → 1,648 m (−22 %); 1,301 m upwind of the pad at apogee against 363 m; main chutes never reached	Yes

6. Conclusion and assumptions:

The final design meets every stated requirement in the Mustafar simulation: apogee 4,010 m, 1.00 kg payload, 1,780 mm length, ≈1,314 N・s total impulse, 4.46 kg mass, no mass overrides, drogue at apogee, payload and main-chute deployment at 2,000 m, and final descent speeds of 5.65 and 6.16 m/s. The Earth-versus-Mustafar tests supported most predictions: lower gravity, thinner air and hotter air each raised apogee, hotter air lowered the Mach number, thinner air sped up the parachute descent, and the side wind cut apogee through weathercocking. The prediction that thinner air would noticeably change stability was not supported.

Assumptions:

1. The parachute and other rocket materials sustain the high temperature.
2. OpenRocket's own atmosphere model is used; its 590 m/s speed of sound at 700 K is higher than an ideal-gas hand estimate (≈530 m/s), so Mach numbers are OpenRocket's. Gravity is set through a plugin extension, and the wind is steady with no gusts.
3. Latitude and Longitude arbitrarily chosen.